

Miguel G. SILVA

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SUMMARY

PhD researcher and applied AI scientist. I take hard modeling problems (e.g., neural surrogates for physics simulation, multi-objective constrained optimization over engineered products) from formulation to production. Background in time-series modeling, pattern discovery, and interpretable ML, with first-author publications in Q1 journals.

WORK EXPERIENCE

2025 – PRESENT	Applied AI Scientist at DAREDATA ENGINEERING - Co-led the design of a production AI pipeline that automates custom cable design for a global automotive cable manufacturer, replacing a 16-week / ~€10k physical prototype-and-lab cycle with a 1-day / ~€1.8k software workflow (~80× faster, 5–10× cheaper per RFP). - Formulated the surrogate-modeling layer: a Graph Neural Network with calibrated uncertainty, trained on multiphysics simulation history, turning hours of physics solves into millisecond inference. - Designed the multi-objective constrained optimization over a mixed continuous/discrete design space (material × geometry × stack), balancing cost and performance under regulatory and manufacturability constraints. Calls the surrogate as a fast oracle, routing survivors to FEM validation. - Designed the agent architecture for the LLM engineer assistant (OpenAI SDK): the specialist agents it can call (generation, simulation, optimization, proposal), the tools each exposes, and how the orchestrator routes intent.
2021 – 2026	Doctoral Researcher at LASIGE & INESC-ID - Analyzed smart meter data from 2,170 households using biclustering, uncovering statistically significant and actionable local water consumption patterns inaccessible to traditional clustering methods. - Developed a scalable methodology to extract population density patterns from mobile phone data (3,743 cells) across Lisbon, uncovering multi-level seasonality and emerging urban trends. - Mined motifs in multivariate time series; built <code>msig</code> to evaluate their statistical significance.
2022 – 2024	Teaching Assistant at IST, UNIVERSITY OF LISBON Invited TA delivering 140+ hours on Information Retrieval (Python) and Database Systems (SQL) to 300+ students; 4.5/5 average rating across three semesters.

EDUCATION

2021 – 2026	PhD in Computer Science <i>Faculty of Sciences University of Lisbon, Portugal</i>
2018 – 2020	MSc in Data Science <i>Faculty of Sciences, University of Lisbon, Portugal</i> GPA: 18 / 20 , THESIS: 19/20

TECHNICAL SKILLS

Languages & Core: Python · SQL · Git · LaTeX

ML / Deep Learning: PyTorch · Scikit-learn · Optuna · Pandas · NumPy

Applied AI: Surrogate Modeling · Multi-Objective Optimization · Time-Series Modeling · Uncertainty Quantification

LLM & Agents: OpenAI SDK · Tool Use · Structured Outputs · Orchestration

Engineering & MLOps: FastAPI · Pydantic · Docker · CI/CD (GitHub Actions) · MLflow · Async Pipelines

PUBLICATIONS

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| 2026 | On Why and How Statistical Significance Criteria Can Guide Multivariate Time Series Motif Analysis
<i>Pattern Recognition Letters</i> PyPI: msig GitHub |
| 2025 | Cutting through the noise: Explaining residuals in multivariate time series with motif analysis
<i>Pattern Recognition</i> GitHub |
| 2025 | The Next Motif: Tapping into Recurrence Dynamics and Precursor Signals to Forecast Events of Interest
<i>Springer (book chapter)</i> GitHub |
| 2024 | A Comprehensive Survey on Biclustering-based Collaborative Filtering
<i>ACM Computing Surveys</i> |
| 2024 | Actionable descriptors of spatiotemporal urban dynamics from large-scale mobile data
<i>Environment and Planning B</i> GitHub |
| 2022 | Water consumption pattern analysis using biclustering: When, why and how
<i>MDPI Water</i> |